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(See 6.9)

MILITARY SPECIFICATION

DEHYDRATOR, LOW PRESSURE AIR, FOR SPECIAL APPLICATIONS

This specification is approved for use within the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification contains requirements for an automatic dehydrator for use in low pressure air systems. The dehydrator shall remove particulate matter, aerosols (see 6.5) and water vapor from influent low pressure air and furnish clean, dry air as specified herein for use in pressurizing radar equipment, waveguide transmission systems, steam propulsion plant control systems, operating automatic combustion control and other special low pressure air applications. The dehydrator shall be designed for both continuous and intermittent operation. The word "dehydrator" denotes the entire complex of equipment throughout this specification.

1.2 Classification. Dehydrators are classified as follows:

1.2.1 Types. The dehydrator shall be of the following types, as specified (see 6.2.1):

Type I - Refrigeration.
Type II - Heater self-reactivating desiccant.

1.2.2 Capacity. The nominal capacity (nominal effluent air flow rate) for the dehydrator shall be of the following classes as specified (see 6.2.1):

Class 3 - 30 standard cubic feet per minute (ft^3/min).
Class 5 - 100 ft^3/min.
Class 7 - Special, nominal capacity as specified (see 6.2.1).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 5523, Department of the Navy, Washington, DC 20362-5101 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and the supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

QQ-C-390	- Copper Alloy Castings (Including Cast Bar).
QQ-C-465	- Copper-Aluminum Alloys (Aluminum Bronze) (Copper Alloy Numbers 606, 614, 630 and 642); Rod, Flat Products with Finished Edges (Flat Wire, Strip, and Bar), Shapes, and Forgings.
QQ-N-281	- Nickel-Copper Alloy Bar, Rod, Plate, Sheet, Strip, Wire, Forgings, and Structural and Special Shaped Sections.
QQ-N-286	- Nickel-Copper-Aluminum Alloy, Wrought (UNS N05500).
TT-E-490	- Enamel, Silicone Alkyd Copolymer, Semigloss (For Exterior and Interior Use).
TT-P-645	- Primer, Paint, Zinc Chromate, Alkyd Type.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-621	- Boxes, Wood, Nailed and Lock-Corner.

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MIL-C-104	- Crates, Wood: Lumber and Plywood Sheathed, Nailed and Bolted.
MIL-P-116	- Preservation, Methods of.
MIL-S-901	- Shock Tests, H.I. (High-Impact); Shipboard Machinery, Equipment and Systems, Requirements for.
MIL-C-915	- Cable and Cord, Electrical, for Shipboard Use, General Specification for.
MIL-E-917	- Electric Power Equipment, Basic Requirements (Naval Shipboard Use).
MIL-F-1183	- Fittings, Pipe, Cast Bronze, Silver-Brazing, General Specification for.
MIL-E-2036	- Enclosures for Electric and Electronic Equipment, Naval Shipboard.
MIL-C-2212	- Controllers, Electric Motor A.C. or D.C., and Associated Switching Devices.
MIL-L-3661	- Lampholder, Indicator Lights, Indicator Light Housings, and Indicator Light Lenses, General Specification for.
MIL-L-3661/65	- Lampholder, Lights, Indicator (Housing), Style LH98.
MIL-M-3971	- Meters, Time Totalizing, Non-Hermetically Sealed, Electrical: General Specification for.

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MILITARY (Continued)

MIL-S-8805 - Switches and Switch Assemblies, Sensitive and Push (Snap Action), General Specification for.
MIL-T-15005 - Tube, 70-30 and 90-10 Copper-Nickel Alloy, Condenser and Heat Exchanger.
MIL-P-15024 - Plates, Tags and Bands for Identification of Equipment.
MIL-P-15024/5 - Plates, Identification.
MIL-S-15291 - Switches, Rotary, Snap Action and Detent/Spring Return Action, General Specification for.
MIL-A-15303 - Audible Signals: Alarms, Bells, Buzzers, Horns, and Sirens, Electronic, Shipboard.
MIL-C-15726 - Copper-Nickel Alloy, Rod, Flat Products (Flat Wire, Strip, Sheet, Bar, and Plate) and Forgings.
MIL-S-16032 - Switches and Detectors, Shipboard Alarm Systems.
MIL-M-17059 - Motors, 60-Cycle, Alternating-Current, Fractional H.P. (Shipboard Use).
MIL-M-17060 - Motors, 60-Hertz, Alternating Current, Integral-Horsepower, Shipboard Use.
MIL-I-18997 - Indicator, Pressure, Panel Mounted or Case Supported, General Specification.
MIL-I-18997/1 - Indicators, Pressure, Circular Dial (3-1/2, 4-1/2 and 8-1/2 Inch Dial Sizes).
MIL-R-19523 - Relays, Control, Naval Shipboard.
MIL-T-19646 - Thermometers, Remote Reading, Self-Indicating Dial, Gas Actuated.
MIL-H-22577 - Heating Elements, Electrical: Cartridge, Strip and Tubular Type.
MIL-I-24144 - Indicators, Frost-Point/Dew-Point, Self-Cooling, Semi-Portable.
MIL-V-24578 - Valves, Globe, Pressure Instrument, Stem Test Connection, Union End.
MIL-M-38510 - Microcircuits, General Specification for.
MIL-C-52950 - Crates, Wood, Open and Covered.

STANDARDS

FEDERAL

FED-STD-H28 - Screw-Thread Standards for Federal Services.

MILITARY

MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-167-1 - Mechanical Vibrations of Shipboard Equipment (Type I - Environmental and Type II - Internally Excited).
MIL-STD-278 - Welding and Casting Standard.
MIL-STD-454 - Standard General Requirements for Electronic Equipment.
MIL-STD-461 - Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference.

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MIL-STD-471 - Maintainability Verification/Demonstration/ Evaluation.
MIL-STD-769 - Thermal Insulation Requirements for Machinery and Piping.
MIL-STD-777 - Schedule of Piping, Valves, Fittings, and Associated Piping Components for Naval Surface Ships.
MIL-STD-1186 - Cushioning, Anchoring, Bracing, Blocking and Waterproofing; with Appropriate Test Methods.
MIL-STD-1367 - Packaging, Handling, Storage, and Transportability Program Requirements (for Systems and Equipments).
MIL-STD-1378 - Requirements for Employing Standard Electronic Modules.
MIL-STD-1389 - Design Requirements for Standard Electronic Modules.
DOD-STD-1399, Section 300 - Interface Standard for Shipboard Systems Electric Power, Alternating Current. (Metric)
MIL-STD-1472 - Human Engineering Design Criteria for Military Systems, Equipment and Facilities.
MIL-STD-1635 - Reliability Growth Testing.

2.1.2 Other Government drawings and publications. The following other Government drawings and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS

MILITARY

803-841499 - Strainer, Bronze, 250 PSI, at 425°F Max.
803-1385850 - Piping, Instruments, Pressure for All Service.
810-2145528 - Fittings and Adaptors, Instrument, Pressure.

PUBLICATIONS

NAVAL SEA SYSTEMS COMMAND (NAVSEA)

0967-LP-597-1011 - Electronic Equipment, Parts Application and Reliability Information Manual for Navy.
0900-LP-001-7000 - Piping System Brazed, Fabrication and Inspection.

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

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AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A 312 - Standard Specification for Seamless and Welded Austenitic Stainless Steel Pipe. (DoD adopted)
B 62 - Standard Specification for Composition Bronze or Ounce Metal Castings. (DoD adopted)
D 3951 - Standard Practice for Commercial Packaging. (DoD adopted)

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.4 and 6.4).

3.2 Reliability. The mean time between failures (MTBF) of the dehydrator assembly shall be not less than 3,000 hours (see 4.7.3.5). A failure shall be any condition or malfunction that requires the dehydrator to be taken off the line for corrective maintenance, or air output that does not meet the minimum dew point, flow capacity, or pressure requirements as specified herein.

3.3 Maintainability. The dehydrator shall be such that all maintenance, both corrective and preventive, can be accomplished at the organizational level with no outside assistance. Preventive and corrective maintenance shall be accomplished by a machinist mate third class with no formal equipment training. Maintainability shall be demonstrated in accordance with 4.7.4.

3.3.1 Preventive maintenance. The mean preventive maintenance time shall be not greater than 10 man-hours per month. The time required to perform any preventive maintenance action shall be not greater than 3 man-hours by a machinist mate third class with no formal equipment training.

3.3.2 Corrective maintenance. The unit shall have a mean time to repair (MTTR) of not more than 3 man-hours. The maximum time to perform any corrective action shall be not greater than 5 man-hours.

3.4 General description. The dehydrator shall be capable of effluent flows from zero to nominal rated flow without requiring manual attention. Unless otherwise specified (see 6.2.1), dehydrators shall be shipboard deck mounted. Each dehydrator shall be housed in a metal cabinet enclosing the top, sides, and back. The cabinet shall protect the internal piping and components of the dehydrator

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from the shipboard environment, for example, dripping salt and fresh water, steam leaks, falling objects such as tools and fasteners, and the intrusion of pipe sections and tool handles as may occur during extensive refit activity in the immediate vicinity.

3.4.1 Type I, refrigerated. The type I dehydrator shall be a regenerative compression refrigerant type intended for continuous noncycling operation at the conditions specified herein. The dehydrator shall consist of a regenerative air-to-air heat exchanger, refrigerant condenser, evaporator, separators, automatic condensate discharge device with manual override, filters, accessories, and instrumentation specified herein. Unless otherwise specified (see 6.2.1), the refrigerant condenser shall be sea water cooled. The unit shall be factory assembled complete, self-contained with the refrigerant equipment dehydrated and charged with the necessary operating quantities of refrigerant and oil. The unit shall operate as specified herein after removal of shipping protection, opening of valves, and connection to services.

3.4.2 Type II, heater self-reactivating desiccant. The type II dehydrator shall be fully automatic. The dehydrator shall consist of desiccant towers, heaters, filters, tower purge assembly, accessories and instruments specified herein. Each tower shall have at least one removable head to facilitate replacement of desiccant and inspection of tower internals. Reactivation purge air flow direction through the desiccant towers shall be the reverse of the dehydrating air flow. Heaterless type II dehydrators shall be prohibited.

3.4.3 Arrangement and accessibility. The dehydrator shall be configured such that operation, maintenance, and repair of the unit can be accomplished from the front of the unit. Piping, wiring, instruments, and other components shall be arranged such that their partial disassembly or removal is not required to provide ready access to components requiring preventive maintenance. Desiccant chambers for a type II dehydrator shall be removable from the front of the dehydrator without disassembly of piping or other components. Test points, test connections, and calibration connections in the dehydrator shall permit easy accessibility without requiring removal or relocation of any components. Threaded captivated fasteners shall be used to facilitate installation and removal of the cabinet panels. Instrument panels shall be mounted as necessary so as not to interfere with the accessibility requirements of this paragraph. This may require double-door arrangements, hinged to the frame, with flexible tubing links and extra length wiring to permit the panels to be swung clear of the maintenance work area within the dehydrator. The panel shall be constructed with easy access for daily operations. Instruments shall be clearly visible to the machine operator.

3.4.4 Physical parameters. The dehydrator shall not exceed the physical parameters specified in table I. Where no size and weight limitations are provided, the physical parameters shall be as specified (see 6.2.1).

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TABLE I. Dehydrator overall dimensions and weights.

Type	Class	Height (inches)	Width (inches)	Depth (inches)	Weight (pounds)
I	5	52	32	34 plus 36 for disassembly	700
II	3	60	36	18 plus 24 for disassembly	700

3.5 Materials. Materials used in the dehydrator shall meet the requirements specified herein. Where no specific requirements are identified, materials used in the dehydrator shall be corrosion-resistant in a marine atmosphere. Air piping shall be copper-nickel or corrosion resisting steel. Materials which are in direct contact with desiccant shall be grade 304 corrosion-resistant steel in accordance with ASTM A 312. Critical parts such as spool pieces, guide bushings, poppets, stems, and so forth, shall be corrosion-resistant materials such as corrosion-resistant steel, corrosion resistant nickel-copper-aluminum alloy in accordance with QQ-N-286, nickel-copper alloy in accordance with QQ-N-281, bronze or equivalent alloys. Rubbing parts in contact with each other shall be non-galling or hardened grades to prevent galling, seizing, or excessive wear. Other materials used in the construction of the dehydrator shall be similarly selected on the basis of corrosion-resistance and anti-galling properties. The use of cast iron shall be permitted only for type I refrigeration compressors.

3.5.1 Asbestos. Material used in the construction of the dehydrator package (components and supply support items) shall be asbestos-free. When specified in the contract or order, a certificate of compliance shall be prepared (see 6.2.2).

3.5.2 Recovered materials. Unless otherwise specified herein, all equipment, material, and articles incorporated in the products covered by this specification shall be new and may be fabricated using materials produced from recovered materials to the maximum extent practicable without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products is allowed under this specification unless otherwise specifically specified.

3.6 Construction. Construction shall be as specified in 3.6.1 through 3.6.25.

3.6.1 Inlet pressure. Construction shall be based on an inlet pressure of 150 pounds per square inch (lb/in^2) for the air and water flow paths. The freon system shall be based on a high side pressure of 250 lb/in^2 and a low side pressure of 40 lb/in^2. To avoid the need for any over-pressure protection devices, tubing and equipment furnished as part of the dehydrator shall be constructed for these pressures.

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3.6.1.1 Proof pressure. The air and water flow paths, and the freon system shall withstand a pressure 1.5 times the respective design pressures (see 4.7.1).

3.6.2 Piping, valves and fittings. Piping, valves and fittings shall be in accordance with MIL-STD-777, except that welded connections shall not be required.

3.6.2.1 Fittings. Pipe thread connections and bite-type fittings shall not be used. Air connections shall be O-ring gasket and retainer ring union connections. The complete union assembly shall be provided with the dehydrator to ease installation. Piping connections shall be arranged in such a manner as to not impair the vibration-isolation properties of absorption-type mounts. Piping shall be securely supported to minimize strain and vibration.

3.6.2.2 Gauge piping. Gauge piping shall be in accordance with Drawing 803-1385850. Gauge piping shall be either corrosion-resisting steel or 70-30 copper nickel.

3.6.2.3 Valves. Valves shall incorporate a soft seating feature to ensure repeated tight shutoff. Construction of valves shall emphasize ruggedness, reliability, simplicity, and ease of maintenance. Solenoid valves shall conform to the electrical requirements of MIL-E-917. Complex valves such as solenoid-triggered, air-operated, multifunction spool valves shall be prohibited. Valves and valve components on type II dehydrators shall withstand a temperature of 550 degrees Fahrenheit ($^{\circ}F$) without affecting valve operation or valve components. The type II dehydrator shall have a solenoid valve at the inlet to ensure that no inlet air shall enter a dehydrator which is not energized. Air check valves shall employ a separate valve body for the valve. The valve shall not be an insert or a sub-component of another fitting or manifold.

3.6.2.4 Threads. Threads shall conform to FED-STD-H28. Where necessary, provisions shall be incorporated to prevent the accidental loosening of threaded parts. Components with pipe threads shall not be used. Welding, brazing, or otherwise altering a fitting which has pipe threads to render it impossible to take down shall also be prohibited. Bolts, nuts, studs, pins, screws, and such other fastenings or fittings shall be of a corrosion-resisting material. Screws exposed in final assembly shall be corrosion-resisting material. Self-tapping screws with machine-screw threads may be used in the cabinet assembly. Sheet-metal screws with sheet-metal threads shall not be used.

3.6.3 Filters. Filters shall have a dial type differential pressure gauge which shall measure the pressure drop across the filter and function as a warning device of a clogged filter (see 3.6.10.4). Maximum filter pressure drop (from clean to changeout) shall not adversely affect the dehydrator pressure drop requirement specified in 3.7.3. Filter elements shall withstand a pressure differential of 75 lb/in^2 without deformation or damage. The unit shall be constructed such that a machinist mate third class can change the cartridge in one housing unassisted in not more than 10 minutes using only common hand tools. Filter elements shall be replaceable. The filter element, filter attachment and mounting, bowl and bowl closure, seals, seal mountings and seal interface surfaces shall be such that they can be assembled only in the correct manner. Filter elements shall be attached to the filter with a positive

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mounting, such as a threaded fastener. For the type I dehydrator, a coalescing filter shall be installed downstream of the evaporator. For the type II dehydrator, a coalescing filter shall be at the inlet to the dehydrator, but downstream of the dehydrator main inlet solenoid valve. The coalescing filter shall be a replaceable cartridge type with associated automatic condensate discharge valve and drain line (see 3.6.16). The filter shall also have a manual drain valve to permit blowdown of the filtering element. The filter cartridge shall perform with 100 percent efficiency the removal of 3-micrometer or larger diameter solids and liquid particles in gas. Flow shall be from inside to outside. Two particulate filters shall be provided for type II dehydrators, one installed at the main flow outlet of each desiccant chamber upstream of any valves. Each particulate filter shall be equipped with a replaceable cartridge, that shall remove particulate matter with not less than 98 percent efficiency for 2-micrometer particles and 100 percent efficiency for 5-micrometer particles. Particulate filter elements and housing shall withstand an air temperature of 550°F without damaging the element or housing.

3.6.4 Interchangeability. In no case shall parts be physically interchangeable or reversible unless such parts are also interchangeable or reversible with regard to function, performance and strength.

3.6.5 Desiccant tower. Each desiccant tower shall be permanently marked with the proper desiccant level. Spring force shall be maintained on the desiccant bed throughout a 4,000-hour bed life and shall adjust for normal bed attrition. Seals, desiccant retainers, the entire desiccant chamber, and all portions of the dehydrator downstream of the desiccant chamber shall not be adversely affected by temperatures of 550°F during the heating portion of the reactivation period. Joints at the interface between towers, fin assemblies, heaters, desiccant retainers, removable end caps and screens shall be of the positive mechanical type. Friction fits such as O-rings shall not be the primary seal. The desiccant tower pressure seals shall be located on the outer surface of the towers for ease of replacement and maintenance.

3.6.5.1 Desiccant (type II). Desiccant shall be activated alumina. The amount of desiccant provided shall permit the dehydrator to conform to the effluent air requirements specified in 3.7.5. The desiccant shall retain all water adsorption qualities after saturation with liquid water and reactivation.

3.6.5.2 Desiccant tower breakthrough capacity. The desiccant tower in the type II dehydrator shall have a minimum breakthrough (see 6.5) capacity time of 150 percent of the actual dehydration period (see 6.5). The moisture adsorption capacity of a desiccant tower shall be not less than 150 percent of the maximum amount of moisture adsorbed during an actual dehydration period under the highest possible moisture influent conditions (80 lb/in² pressure and saturated at 105°F) and maximum ambient temperature (see 3.7.1). Reserve desiccant capacity shall be demonstrated during the breakthrough test (see 4.7.3.1).

3.6.6 Dew point monitor and alarm. A moisture (dew point) monitor and alarm shall be provided for the type II dehydrator to give a "go no-go" indication that the dehydrator is functioning properly. It shall continuously monitor the moisture level of the effluent air and shall cause a visual and audible alarm to energize on the dehydrator when the dew point rises to a preset value. The alarm actuation value shall be a dew point of air at minus 20 ± 7°F at a pressure of 80 lb/in². The audible alarm shall be a buzzer in accordance with MIL-A-15303

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and shall be provided with a cutout switch to permit silencing of the audible alarm while the visual alarm remains actuated. Dehydrator shall be such that remote/additional alarms may be installed. The monitoring and alarm system shall be constructed for calibration verification and calibration at the instrument panel without removal of any of the components from the dehydrator. The moisture alarm system shall simplify operation and maintenance. The dehydrator shall be provided with a contact closure for extension of audible and visual alarms. An in-line 6800-ohm, 2-watt resistor shall permit switchboard supervision to demonstrate that the circuit is complete. Electromagnetic interference provisions shall be incorporated in the dehydrator in accordance with MIL-STD-461.

3.6.6.1 Calibration of moisture alarm. Calibration shall be performed by mixing dry air with undried air. A flowmeter to measure undried air flow shall be permanently affixed to the dehydrator along with the necessary valves to regulate the mixing of the two air streams, isolate them from the dry air outlet, and permit readout of the resultant mixture from sampling outlet with an auxiliary moisture instrument in accordance with MIL-I-24144. An auxiliary moisture instrument shall be connected to the sampling port. Air of the desired dew point shall then be created using the flow meter and valves. The moisture alarm shall then be calibrated and tested. A caution plate indicating that the flowmeter shall only be used when testing the dew point sensor shall be located as close to the flowmeter as practicable.

3.6.6.2 Dew point alarm requirements. A cut-out switch shall silence the audible alarm while the visual alarm remains energized. The alarm actuation point shall be adjustable without special tools over a dew point range not less than minus 60 to 0°F at a pressure of 80 lb/in^2. Moisture alarm system operation shall be repeatable and shall be within the specified accuracy limits for a minimum interval of 1 year; that is, the alarm system shall not require servicing or calibration on a frequency schedule of less than 1 year. Furthermore, the alarm system shall include a wide scale range to assure repeatability of adjustment within the specified accuracy limits of the system; that is, having determined a set point for a given moisture level, the operating personnel can vary the adjustment and subsequently return the adjustment to the formerly determined set point (without the aid of test equipment) with the assurance that the alarm system will function within the specified accuracy limitations. The adjustment position corresponding to the specified set point of the alarm system (equivalent to a dew point of minus 20°F at a pressure of 80 lb/in^2) shall be identified on each piece of equipment delivered to the Government. Performance of the moisture alarm system shall not deteriorate as a result of prolonged exposure (100 hours or more) of the sensor to moisture levels of 95 percent relative humidity. Furthermore, the performance shall not deteriorate as a result of short exposures (1 hour or less) of the sensor to water. The alarms shall energize within 25 seconds when the dew point rises to minus 20°F at 80 lb/in^2. The alarms shall de-energize within 25 seconds when the dew point drops below minus 20°F at 80 lb/in^2. An electrically isolated circuit shall be provided with normally open contacts which shall close when dew point alarms are energized and shall reopen when the alarms de-energize. The circuit shall terminate at a set of contacts in a connection box.

3.6.7 Purge air systems. The type II dehydrator shall be provided with a means to regulate the purge flow rate. A variable area purge air flow indicator calibrated in ft^3/min shall be provided. Purge pressure shall be continuously

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monitored using a dedicated pressure gauge for each tower. The required purge flow rates shall be marked or banded to the purge flow indicator. A purge flow switch shall be provided (see 3.6.9.2). Purge flow switch operation shall not be affected by the presence of desiccant dust in the switch.

3.6.8 Electrical design. The dehydrator shall meet the requirements specified in MIL-STD-461 and MIL-E-917. The type I dehydrator shall be constructed for 440 volts alternating current (Vac), 3-phase, 60 hertz (Hz). The type II dehydrator shall be constructed for 115 Vac, single-phase, 60 Hz. Both type I and type II dehydrators shall use type I power in accordance with DOD-STD-1399, section 300. Upon loss of and after restoration of power, the same sequence of events shall be automatically restored as occurred prior to the loss. Compressor and fan motors shall be service A with class B or F insulation in accordance with MIL-M-17059 (fractional horsepower (hp)) or MIL-M-17060 (integral hp). Motors shall withstand a total maximum temperature limit of 248°F with limiting temperature rises as specified in MIL-M-17059 or MIL-M-17060, as applicable, for class B insulated motors in a 122°F ambient. Relays shall be in accordance with MIL-R-19523. Type II dehydrators shall be provided with a rotary on-off switch conforming to MIL-S-15291. Type I dehydrators shall be provided with a motor controller and overload relay in accordance with MIL-C-2212. Electrical and electronic components shall be derated in accordance with NAVSEA 0967-LP-597-1011. Heating elements shall be wired with electrical disconnects to permit maintenance and repair without disconnecting the wiring in the control box.

3.6.8.1 Electrical connectors. Electrical connectors shall be in accordance with requirement 10 of MIL-STD-454. Connectors shall be provided with mating parts. Alarm circuit connectors shall be accessible in order to permit terminal connections for remote alarms and independent circuit closure contacts. The power feed line shall be constructed for "hard wired" shipboard installation.

3.6.8.2 Cabling and components. Cabling and electrical components in the dehydrator shall be protected against transients from all sources. The profile of the transients to be protected against shall include a maximum voltage peak seven times normal line voltage having a duration of 5 microseconds. Transients shall not cause damage or inaccurate readings of electrical instruments exceeding the allowable tolerance for that instrument.

3.6.8.3 Wiring and enclosures. Electrical cable shall be in accordance with MIL-C-915. Electrical enclosures shall be watertight and shall be in accordance with MIL-E-2036. Wiring shall be neatly laid out and attached in an orderly fashion. Wiring shall be clearly identified with wire numbers.

3.6.8.4 Terminals and terminal boards. Terminals and terminal boards shall be in accordance with MIL-C-2212.

3.6.8.5 Motor controllers. Motor controllers and associated pressure switches shall be in accordance with MIL-C-2212. Remotely located pressure switches shall be in watertight enclosures. The controllers shall have the following characteristics:

(a)	Enclosure	Dripproof
(b)	Operation	Magnetic

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(c)	Ambient temperature	122°F
(d)	Insulation	Class B
(e)	Control function	Motor starting
(f)	Type	Across the line full voltage nonreversing
(g)	Protection	Low voltage release Overload relay
(h)	Duty	Same as motor

3.6.9 Controls. Controls shall be installed which shall permit electrically tying together several components of a shipboard air system. Switches used to actuate alarms shall be in accordance with MIL-S-16032. Visual alarms shall be in accordance with MIL-S-16032. Visual alarms shall be lamps (see 3.6.10.3). Audible alarms shall be buzzers in accordance with MIL-A-15303 and provided with a cut-out switch to silence the audible alarm while the visual alarm remains actuated.

3.6.9.1 Type I. The dehydrator shall be provided with audible and visual temperature alarms. Evaporator outlet temperatures greater than 45°F and less than 34°F shall actuate these alarms. Dehydrator shall be interlocked with the associated compressor so that the compressor shall shut down upon shutdown of, or loss of electrical power to the dehydrator. A time delay of 3 minutes maximum on the dehydrator shall temporarily prevent energizing the high temperature alarm during dehydrator startup. Controls shall be provided on the panel to override the compressor/dehydrator interlock while receiving compressed air from a shore source. The remote temperature alarm contacts shall supply a 115 Vac signal during the alarm mode and the remote time delay contacts shall supply a 115 Vac signal upon termination of the delay period.

3.6.9.2 Type II. The dew point of the effluent air shall be continuously monitored (see 3.6.6) so that a dew point above the preset level shall energize the audible and visual alarm (see 3.6.6.1 and 3.6.6.2). A purge flow switch shall energize audible and visual alarms and cut off power to the heater in use and the repeat cycle program timer in the event that purge air flow is less than the minimum required value to ensure proper reactivation of the desiccant. The purge flow switch shall restore power to the heater and repeat cycle program timer when the minimum purge air flow is restored. The heaters (see 3.6.14) shall incorporate controls to maintain the desiccant at 450°F during the heating portion of the reactivation period. In addition, the dehydrator shall be provided with a means to cut off electrical power to the heater and repeat cycle program timer and energize an audible and visual alarm when any desiccant exceeds 550°F. This high temperature cutoff and its associated sensor shall be separate from the heater controls. Means shall be provided to limit the rate of pressurization and depressurization during tower switchover. The rate of pressurization and depressurization shall not exceed 5 lb/in^2 per second without disrupting outlet air flow. Positive and tamperproof means shall be provided, either mechanically or electronically by timing and sequential operators or control systems, to prevent operation of a dehydration chamber (for dehydrating inlet air) that has not immediately previously passed through a reactivation cycle to remove water from the dessicant and has not cooled down to 120°F (49 degrees Celsius (°C)) maximum. The operator shall be warned by a flashing light or other visual means that an automatic interlock has engaged because of an improper operating sequence.

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3.6.10 Instrumentation. Instrumentation shall be as specified in 3.6.10.1 through 3.6.10.4.

3.6.10.1 Pressure indicators. Pressure indicators shall be in accordance with MIL-I-18997 and MIL-I-18997/1, and shall have the following characteristics:

(a) Dial size: 3-1/2 inches.
(b) Range: 0 to 200 lb/in^2.
(c) Minor graduations: 5 lb/in^2.
(d) Design: Simplex.
(e) Operating pressure pointers shall be in accordance with MIL-I-18997. One shall be set at the minimum rated operating pressure and one at the maximum. Permanent markings on the indicator face may be substituted. The markings shall clearly designate the rated operating pressure range and the remainder of the face shall be marked as a danger zone.
(f) Indicators shall be installed in accordance with Drawings 603-1385650 and 810-2145528. Pressure gauges shall be such that they may be calibrated without removal from the instrument panel. Gauge calibration valves shall be provided in accordance with MIL-V-24578.

3.6.10.2 Temperature indicators. Temperature indicators shall have an accuracy of plus or minus 3 percent of the full scale reading and shall be non-mercuric.

3.6.10.2.1 Type I dehydrator. A direct sensing temperature indicator in accordance with MIL-T-19646, class A, shall be provided. Its sensor shall indicate evaporator outlet air temperature, and shall not be adversely affected by zero air flow or by high ambient temperatures. Its face shall have two additional pointers, set to indicate minimum and maximum operating temperatures of the dehydrator (34°F and 45°F). Permanent markings on the indicator face may be substituted. The markings shall clearly designate the rated operating temperature and the remainder of the face shall be marked as a danger zone.

3.6.10.2.2 Type II dehydrator. A separate sensing temperature indicator shall be installed on each tower. Both temperature indicators shall function whenever the dehydrator is in operation.

3.6.10.3 Lamps and lamp test switches. Lamp test switches shall be in accordance with MIL-S-8805 and shall be the push (snap action) type. Lamps shall be style LH98 in accordance with MIL-L-3661 and MIL-L-3661/65. Alarm lamps shall flash continuously when actuated.

3.6.10.4 Differential pressure indicators. Differential pressure indicators shall be provided for all filters. The differential pressure at which filter change is required shall be marked on the indicating face of the differential pressure gauge or on the identification plate of the gauge. The filter gauge shall read zero under conditions of zero air flow and proportionately higher positive pressure representative of the filter media condition and air flow. Both the gauge and its identification plate shall be clearly visible to the operator while standing in front of the dehydrator and observing the instrument panel. Materials and testing shall be in accordance with MIL-I-18997.

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3.6.11 Instrument panels. Each dehydrator shall be provided with an instrument panel for centrally locating all monitoring devices.

3.6.11.1 Type I dehydrator instrument panel. The type I dehydrator instrumentation shall consist of the following:

(a) Inlet pressure indicators (lb/in^2).
(b) Lamp test switch.
(c) Power-on lamp (white).
(d) Temperature alarm lamp (red).
(e) Time totalizing meter in accordance with MIL-M-3971 which shall indicate dehydrator operational time.
(f) Temperature indicator.
(g) Filter differential pressure gauges (which shall be visible while observing the instrument panel).
(h) Refrigerant high/low pressure cut-out indicator.
(i) Refrigerant compressor suction pressure gauge.
(j) Refrigerant compressor discharge pressure gauge.
(k) Share air compressor-dehydrator override switch.

3.6.11.2 Type II dehydrator instrument panel. The type II dehydrator instrumentation shall consist of the following:

(a) Inlet and outlet pressure indicators.
(b) Lamp test switch.
(c) Lamps for each tower to indicate:

    (1) When tower is on dehydration service (green).
    (2) When tower is on reactivation service (blue).
    (3) When heater is energized. Light shall indicate actual power consumption (amber).
    (4) When dehydrator effluent dew point rises above dew point alarm set point (red).

(d) Desiccant tower temperature indicator.
(e) Purge flow indicator (accuracy shall be plus or minus 5 percent of full scale reading).
(f) Moisture alarm (visual alarm lamp shall be located on instrument panel, audible alarm need not be mounted at instrument panel).
(g) Power-on lamp (white).
(h) Time totalizing meter in accordance with MIL-M-3971, except that it shall have a total capacity of 99,999 hours with no tenths register necessary. Meter shall indicate dehydrator operational time.
(i) A visual indicator (which shows the phase of dehydrator operation).
(j) Filter pressure differential gauges.
(k) A direct measuring outlet flowmeter indicating 0 to 150 percent of dehydrator rated nominal effluent flow rate installed with a bypass line. The meter shall be of the rodguided rotameter type.
(l) A flow-limiting device in the dehydrator discharge line to limit air flow to 115 percent of nominal capacity with failsafe features and low pressure drop at nominal capacity.

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(m) Low purge flow lamp to activate and flash during low flow purge conditions.

(n) Dehydrator tower high temperature alarm light.

3.6.12. Dew point sample connection. Each dehydrator shall have a dew point sample connection installed in the dehydrator effluent air line. This sample connection shall be located in the front of the dehydrator and shall be a self-sealing quick disconnect with a plug type. The plug shall be attached to the instrument with a cable to insure against loss. The connector shall be the quick disconnect type with "E" packer seal made from a compound intended for the specified temperature and pressure ranges. This connection shall permit sampling with a portable instrument in accordance with MIL-I-24144.

3.6.13 Repeat cycle program timers. Repeat cycle program timers shall be either electromechanical or electronic. The timer shall allow continuous unattended operation of the type II dehydrator.

3.6.13.1 Electro-mechanical repeat cycle program timer. The electro-mechanical type repeat cycle program timer shall:

(a) Have a normal switch life of 50,000 operations minimum, at rated voltage and current.
(b) Have a rated motor life of not less than 60,000 hours.
(c) Be automatically self-starting when power is applied.
(d) Indicate visually the different phases of the dehydrator cycle underway at all times. The indicator shall be color coded and coordinated to the indicator lights and reactivation tower temperature indicators so that the operator may quickly identify an abnormal condition. The display shall be a drum with circumferential bands for the functions indicated. Orientation shall be such that the axis of the drum is parallel to and adjacent to the row of indicator lights for maximum correlation.
(e) Not employ plastic materials for gears, cams, or shafts. Gears and cams shall be made of non-corrosive metal. Cams shall be mechanically fixed to the driveshaft in the proper sequence. Timers shall not be damaged by manual rotation in either direction. Positive and tamper-proof means shall be provided to preclude manual operation or manipulation of the entire repeat cycle program timer including the motor and indicator drum.

3.6.13.2 Electronic repeat cycle program timers. Use of microelectronic technology shall be considered in the construction of timers and equipment and maximum use shall be made of standard microelectronic devices (see MIL-STD-1378 and MIL-STD-1389). Microelectronic devices shall conform to MIL-M-38510 and MIL-STD-454. Electronic repeat cycle program timers shall:

(a) Be designed for 110V, single or 3-phase, 60 Hz power.
(b) Have an operating life of not less than 60,000 hours.
(c) Be automatically self-starting when power is applied.
(d) Visually indicate the different phases of the dehydrator operational cycle at all times. The indicator shall be color coded and coordinated to the indicator lights.

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(e) Have a positive and tamperproof means to preclude manual operation or manipulation of the entire repeat cycle program timer (to prevent the operator from interfering with its normal sequential operation).

3.6.14 Heaters. Heating elements shall be of type III (tubular) construction in accordance with MIL-H-22577 and shall not make direct contact with the desiccant. The heater elements shall be removable from the desiccant towers. The heater elements during removal shall not disturb the desiccant packing. External terminals shall be provided to verify satisfactory heater element resistance or amperage values. The heaters shall be controlled as specified in 3.6.9.2. The heaters shall not be subject to a reduction of insulation resistance below the specified limits in accordance with MIL-H-22577 when exposed to a saturated airstream (see 6.5) at the pressures specified in 3.6.1.

3.6.15 Refrigeration unit. The refrigerant for the condensing unit shall be dichlorodifluoromethane (R-12). The unit shall contain a compressor, condenser, controls, and accessories. The compressor shall be a semi-hermetic, two cylinder unit, driven by a 3-phase motor. The evaporating temperatures shall be in the 34 to 45°F temperature range with a capacity of approximately 12,000 British thermal units (Btu) per hour (1 ton) of refrigeration at the rated loading. A continuously running cooling fan shall give additional cooling to the compressor while operating in high ambient temperatures. Unless otherwise specified (see 6.2.1), the condenser shall be sea water cooled.

3.6.15.1 Air cooled refrigerant condenser. When specified (see 6.2.1), the condenser shall be forced-air cooled and constructed of finned tubes. Fins shall be soldered, welded, or mechanically bonded to the tubes. The condenser shall have sufficient cooling surface area to permit dehydrator operation in an air ambient temperature of 120°F while meeting all the performance requirements specified herein.

3.6.15.2 Sea water cooled refrigerant condenser. The condenser shall be water cooled and shall use sea water at a maximum inlet temperature of 95°F, minimum inlet temperature of 34°F, and a 110°F gas temperature. The refrigerant condenser shall use a minimum fouling factor of 0.0005 $(hr\text{-}ft^2\text{-}°F)/Btu$. Water velocities in the condenser shall not exceed 6 feet per second (ft/sec). There shall be no compressor motor overload with sea water entering the condenser at 100°F. The condenser shall be a type in which the sea water tubes can be mechanically cleaned and shall be installed so that the tubes may be cleaned without removing the condenser from the equipment. Condenser components shall be restricted to materials conforming to MIL-C-15726 for 90-10 copper-nickel, QQ-C-465 for aluminum bronze, and QQ-C-390 for alloys C90300 and C92200. Condenser tubes shall conform to MIL-T-15005. The condenser shall purge both the refrigerant and water circuits and drain the water circuit. Condenser gaskets shall be neoprene or equal.

3.6.15.3 Water regulating valves and strainer. A sea water-regulating valve shall be installed immediately downstream of each condenser. The valve shall be of the direct-acting or pilot-controlled type actuated by condenser gas pressure to modulate the flow of water required for the condenser. The valve shall regulate the water flow from shutoff to required capacity within a maximum operating gas pressure rise of 40 lb/in^2. The valve shall be adjustable within an operating refrigerant gas pressure range of 90 to 140 lb/in^2 for R-12. The

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capacity of the water-regulating valve shall at least equal the water requirement of the condenser, based on available inlet water pressure range of 20 to 175 lb/in^2. The valve and all its components shall be constructed of nonferrous or corrosion-resisting material. The valve body shall be valve bronze conforming to QQ-C-390, alloy C92200. The internal metal parts subject to corrosion or erosion shall be made of nickel-copper conforming to QQ-N-281. The valve shall have union ends conforming to MIL-F-1183 on the cooling water inlet and outlet. The valve shall prevent the entry of water into the refrigerant system in the event of valve failure. A strainer in accordance with Drawing 803-841499 shall protect the valve. Cooling water piping adapters may be nickel-copper alloy conforming to QQ-N-281. Cooling water piping fittings may be bronze conforming to ASTM B 62. The strainer shall be not less than 3/4-inch nominal pipe size (nps). The water-regulating valve shall be located within the cabinet so that no refrigerant piping or connections are outside the cabinet enclosure.

3.6.15.4 Refrigerant dryer. A disposable dryer shall be installed in the refrigerant circuit. The dryer shall be equipped with an auxiliary screen or other protective means at the dryer outlet to prevent passage of the dehydrating agent in the event of rupture of the cartridge screen outlet.

3.6.15.5 Hot gas bypass valve. On the type I dehydrator a hot gas bypass valve shall be installed to preclude evaporator freeze-up during low load conditions. The valve shall be adjustable from 0 to 40 lb/in^2 and shall be externally actuated. The discharge of the bypass valve shall be connected to a point between the refrigerant expansion valve outlet and the evaporator inlet.

3.6.15.6 Refrigerant thermal expansion valve. A fine mesh strainer shall be installed upstream from the expansion valve. Strainers for installation in refrigerant lines shall be constructed of copper or brass shell and shall be angle-type arranged for screen removal without disconnection of piping. Flange ring and flange cover plate shall be bronze castings or bronze forgings constructed to withstand distortion caused by uneven drawing up of bolts. The gasket shall be fully retained. A minimum of four bolts and nuts constructed of silicon bronze shall be provided for the flange. The strainer housing shall be constructed for a minimum working pressure of 250 lb/in^2. Screen cartridges shall be constructed to retain foreign matter on the inside of the screen basket, and shall be arranged for adequate clearance between the outside of the screen and the inside of the shell. The screen cartridge shall be positively located and supported. The screen cartridge shall be 100-mesh and shall be supported and mechanically strengthened by an additional screen of coarser mesh. The cartridge assembly shall withstand without rupture or distortion a pressure drop of 125 lb/in^2 in the event of complete stoppage.

3.6.15.7 Evaporator. The refrigerant-to-air heat exchanger (refrigerant evaporator) on a type I dehydrator shall be of copper or other materials approved for tubing (see 3.5). Construction shall not be subject to fouling. Fins, extended surface, "porobond", "pin-fin", fluted-surface, and similar heat exchange devices shall be prohibited. Tube-in-tube or tube-in-shell evaporators shall be preferred. The evaporator shall use a minimum fouling factor of 0.0002 (hr-ft^2-°F)/Btu. The evaporator shall be provided with a condensate device (see 3.6.16) constructed from corrosion-resistant materials.

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3.6.15.8 Refrigerant pressure cut-out switch. A totally enclosed high and low pressure switch with a manual reset shall be required on type I dehydrators. This switch shall cut power to the dehydrator in the event that refrigerant pressure at the compressor discharge rises above 250 lb/in^2 or falls below 18 lb/in^2. This switch shall prevent damage to the dehydrator in the event of loss of cooling water or refrigerant. An instrument panel-mounted alarm lamp shall light when the dehydrator shuts down due to this cutout switch. The lamp shall be equipped with a lamp test switch to check lamp function.

3.6.15.9 Sight glass. A sight glass shall be installed in the refrigerant line between the condenser and expansion valve.

3.6.16 Condensate discharge devices. The condensate discharge devices for the type I and type II filters and the type I evaporator shall be constructed of corrosion-resistant materials, automatic in operation, float actuated, pneumatic power operated (as with a pilot actuated valve), positive acting (as utilizing a magnetic latch or overcenter levers), and fitted with a manual drain valve.

3.6.17 Regenerative air pre-cooler. The type I dehydrator shall have a regenerative air pre-cooler to cool the influent air as it enters the dehydrator and before it reaches the evaporator. The cooling fluid shall be the dehydrator air prior to leaving the dehydrator. At maximum load conditions, the warmed dehydrated air leaving the pre-cooler shall have been raised to within 15°F or less of the wet air entering the pre-cooler. Construction shall not be subject to fouling. Fins, extended surface, "porobond", "pin-fin", fluted surface, and similar heat exchange devices shall be prohibited. The pre-cooler shall be tube-in-tube or tube-in-shell. Material shall be copper or other materials approved for tubing.

3.6.18 Vibration eliminators. The type I dehydrator refrigeration compressor shall be fitted with a vibration eliminator (flexible connection) on both the suction and discharge lines.

3.6.19 Drains. Each drain line shall terminate at a common open funnel drain. Funnel drain shall be visible from the front of the dehydrator for verification of individual drain operation. The common drain shall terminate at the perimeter of the dehydrator with a union, ready for hard piping to the ship's waste-water drain collecting system. The common drain shall be 0.375 inch nps.

3.6.20 Flow limiting devices. A flow limiting device shall be installed in the type II dehydrator discharge line to limit air flow to 115 percent of nominal capacity. The flow limiting device shall have fail-safe features and low pressure drop at nominal capacity.

3.6.21 Painting. Surfaces to be painted shall be thoroughly cleaned and free from oil, grease, mill scale, rust and loose weld beads. External non-working ferrous and aluminum surfaces shall be painted with one coat of zinc chromate primer in accordance with TT-P-645 and two coats of haze-gray paint in accordance with TT-E-490.

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3.6.22 Welding. Welding and allied processes shall be in accordance with MIL-STD-278. Brazing shall be in accordance with NAVSEA 0900-LP-001-7000.

3.6.23 Safety. Placement of devices requiring attention during maintenance or repair shall be such that potential danger to the operator due to high voltage, pressure, high heat, or moving parts shall be precluded. The desiccant towers shall be protected in such a manner to prevent injury to operating personnel.

3.6.24 Human engineering. Dehydrator shall be constructed in accordance with MIL-STD-1472 to preclude or minimize the possibility of failure through improper operation or excessive and complicated maintenance procedures.

3.6.25 Condensation (type I). On the type I dehydrator, any surface on which condensation forms during the maximum load test (see 6.5) in a 105°F ambient wet bulb atmosphere shall be insulated in accordance with MIL-STD-769 to prevent formation of condensation. Insulation shall be removable and reusable in areas requiring routine scheduled maintenance and repair.

3.7 Performance requirements. Performance requirements shall be as specified in 3.7.1 through 3.7.6.

3.7.1 Ambient temperature. The dehydrator shall operate, and meet all the performance requirements specified herein, at any ambient temperature within the range of 50 to 120°F.

3.7.2 Shock and vibration. The dehydrator shall be resistant to mechanical shock and vibration as encountered on board ship. The dehydrator shall pass the shock and vibration tests specified in 4.7.7.

3.7.3 Pressure drop. At rated capacity, the air side pressure drop across the complete dehydrator shall not exceed 5 lb/in^2 for type I and 13 lb/in^2 for type II.

3.7.4 Purge air (type II). The outlet air flow as a percentage of the inlet air flow for the type II dehydrator shall be not less than 90 percent.

3.7.5 Performance parameters. Performance parameters for dehydrators shall be as specified in table II. Sea water cooled type I dehydrators shall be rated with sea water entering the condenser at 95°F. Performance parameters for class 7 dehydrator shall be as specified (see 6.2.1).

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TABLE II. Performance parameters.

Air flow	Maximum temperature (°F)	Pressure (lb/in^2)	Maximum moisture content (lb H_2O/lb dry air)
Type I (Refrigeration)			
Influent Effluent	125 [1](see 3.6.17)	80-150 75-150	0.015 0.00085 (40°F dew point) (see 6.5) at 80 lb/in^2
Type II (Heater self-reactivating desiccant)			
Influent Effluent	105 [2]160	80-150 67-150	0.008 0.000015 (minus 40°F dew point) at 80 lb/in^2

[1] Higher transients shall be permitted when equipment is heat soaked by ambient temperatures greater than 110°F.
[2] Magnitude of transients at changeover shall be minimized and shall not exceed 190°F.

3.7.6 Inclined operation. Type I and type II dehydrators shall meet all the performance requirements specified herein when permanently inclined to 15 degrees (see 4.7.6).

3.8 Marking. Marking shall be as specified in 3.8.1 through 3.8.4.

3.8.1 Identification plates. Plates shall be installed in accordance with MIL-P-15024 and MIL-P-15024/5 and shall be constructed for severe service. Where required, identification plates and information plates shall be mechanically attached to the part of machinery or equipment that will not ordinarily be renewed during its normal service life. Plates shall be located where they will be readily visible. Identification plates and information plates shall be made of brass, nickel-copper alloy, corrosion-resistant steel, or anodized aluminum. Markings shall be either etched, stamped, engraved, metal photographed on anodized aluminum or cast in such a manner as to produce permanent markings. Etchings, engravings, or stampings shall be not more than 0.003 inch deep. Characters on plates shall be raised to at least 0.03 inch. Engraved, stamped or direct etched markings shall be filled with black enamel or lacquer.

3.8.1.1 Dehydrator identification plate markings. Identification plates shall be secured to equipment with corrosion-resistant fasteners. Fasteners used to secure the plates shall not penetrate the frame or the end brackets. Non-metallic identification plates shall not be acceptable. The identification plates shall contain data as follows:

(a) Manufacturer's name.
(b) Manufacturer's model or type and size.
(c) Classification.

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(d) Manufacturer's serial number.
(e) Date of manufacture.
(f) Characteristics.

(1) Capacity.
(2) Inlet pressure (lb/in^2).
(3) Electrical data (for example, voltage).
(4) Dewpoint and corresponding pressure, and temperature.
(5) Moisture content of influent and effluent air.
(6) Any other pertinent characteristics.

(g) Contract or order number (and item number of multiple unit orders).
(h) National stock number. Space for number shall be provided even though number may not be assigned.
(i) Section for inspector's stamp.

Each driving unit and each accessory unit shall have an identification plate and data elements in accordance with the applicable equipment specification.

3.8.2 Piping connections. Inlet and outlet union-ended air connection piping and sizes shall be as specified (see 6.2.1).

3.8.3 Electrical and mechanical controls. Electrical and mechanical controls shall be marked as specified herein.

3.8.4 Operation and safety instruction plate. An operating and safety instruction plate shall give the operator a clear, concise, step-by-step procedure for startup and operation of the dehydrator. The instruction plate shall include:

(a) Operation of the repeat cycle program timer and control valves in conjunction with the various indicator lamps.
(b) Schematic flow diagram with the parts identified therein.
(c) Safety precautions required for safe operation and maintenance.
(d) Type of desiccant, desiccant National stock number and useful desiccant operating life (type II).
(e) Applicable NAVSEA technical manual number and number of this specification.
(f) Step-by-step instructions for start-up, operation, and shutdown of the dehydrator.

3.9 Drawings. When specified in the contract or order, drawings shall be prepared (see 6.2.2).

3.10 Imaged aperture/tabulating cards. When specified in the contract or order, imaged aperture/tabulating cards shall be prepared (see 6.2.2).

3.11 Failure mode, effects, and criticality analysis (FMECA). When specified in the contract or order, a FMECA shall be prepared for use during first article testing (see 6.2.2).

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3.12 Failure/malfunction report, contractor engineering and technical services (CETS) report, and maintainability demonstration report. When specified in the contract or order, a failure/malfunction report, CETS report, and maintainability demonstration report shall be prepared (see 6.2.2).

3.13 Workmanship. Dehydrator units shall be constructed, assembled and finished to assure quality equipment and neat appearance that is free from imperfections that will affect durability, operability, serviceability, and safety. The metal portion of the units shall be free from fins, burrs, sharp and rough edges.

3.14 Technical manuals. Technical manuals (including technical repair standards) shall be prepared and delivered as specified in the Technical Manual Contract Requirements (TMCR) and contract or order (see 6.2.1).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Inspection system. When specified in the contract or order an inspection system program plan shall be prepared (see 6.2.2).

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.4).
(b) Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified (see 6.2.1), all inspections shall be performed in accordance with the test conditions specified in the reference documents.

4.4 First article inspection and testing. The first dehydrator of each type and capacity (see 1.2.1 and 1.2.2) representing production equipment in

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all respects shall be examined and tested to determine conformance to this specification. The first article inspection shall consist of the examination and tests specified herein and shall be conducted in the following order:

(a) Visual and dimensional examination (see 4.6).
(b) Hydrostatic pressure test (see 4.7.1).
(c) Pressure decay test (see 4.7.2).
(d) Performance tests (see 4.7.3).
(e) Maintainability demonstration (see 4.7.4).
(f) Solid contaminent determination (see 4.7.5).
(g) Inclined operation (see 4.7.6).
(h) Shock and vibration (see 4.7.7).
(i) Operating test (see 4.7.8).

The dehydrator to be inspected shall be sent to the U.S. Naval Ship System Engineering Station (NAVSSES), Philadelphia, PA for completion of tests and examinations (a) through (g). Upon successful completion of tests (a) through (g) the dehydrator will be returned to the contractor for completion of tests (h) and (i). Test (i) shall be performed only upon successful completion of test (h). The contractor shall correct all failures, malfunctions or problems resulting from all first article tests and examinations including those performed by the Government. Failures during tests (a) through (g) will require, at the Government's option, a retest by the Government. Failures during tests (h) or (i) shall require a repetition of the failed test by the contractor.

4.4.1 First article reports. When specified in the contract or order, a notification of tests, first article test report, and test procedures shall be prepared for those tests not conducted at NAVSSES (see 6.2.2).

4.5 Quality conformance inspection. Quality conformance inspection shall be classified as follows and shall be performed on each dehydrator. Shutdown, alteration to equipment, or adjustment of any kind prior to completion of each test shall void the inspection unless specifically required by the inspection procedure. Order of tests may be designated by the contractor:

(a) Visual and dimensional examination (see 4.6).
(b) Pressure decay (see 4.7.2).
(c) Moisture alarm test (see 4.7.3.6).
(d) Hydrostatic pressure test (see 4.7.1).
(e) Refrigerant leakage test (see 4.7.3.2).
(f) Operating test (see 4.7.8).

4.5.1 Test procedures. When specified in the contract or order, quality conformance test procedures shall be prepared (see 6.2.2).

4.6 Visual and dimensional examination. Each dehydrator shall be examined to determine conformance to this specification. Equipment shall be surface-examined including visual examination for defects, workmanship, dimensions, and any other requirements not involving tests.

4.7 Tests. Tests shall be conducted in accordance with 4.7.1 through 4.7.8.2.

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4.7.1 Proof pressure test. The air and water flow paths, and the freon system shall be tested at 1-1/2 times the respective design pressures for 1/2 hour. The air side of each dehydrator shall be hydrostatically tested with clean fresh water at a pressure of 225 lb/in^2. The water side of the type I dehydrator shall be hydrostatically tested at a pressure of 225 lb/in^2. The air and water proof pressure tests shall be performed with no freon side pressure. The freon system shall be tested to a high side pressure of 375 lb/in^2 and a low side pressure of 60 lb/in^2 with no air or water side pressure. Any leakage, porosity, or permanent deformation shall be cause for rejection.

4.7.2 Pressure decay test. The air side of each dehydrator shall be pneumatically tested for pressure drop at the nominal operating pressure (no flow through dehydrator) for 1 hour. Assuming temperature corrections measured by a surface measuring instrument, the allowable pressure drop shall not exceed 1 lb/in^2 during the 1-hour test period.

4.7.3 Performance tests. Performance shall be tested in accordance with 4.7.3.1 through 4.7.3.6. The reliability test (see 4.7.3.5) shall be the first test performed. Fresh (tap) water may be used when testing water cooled units. Calibrated flow meters with an error not exceeding 5 percent shall be used. The calibration data, name of manufacturer, and model number for the flow meters shall be recorded. Dew point measurements of the effluent air shall be recorded hourly to establish dehydrator performance. Dew point measurements during testing shall also be taken within 10 minutes prior to the completion of the dehydration period. Dew point error shall not exceed 15 percent. Specified flow rate shall be maintained within minus 3 percent and plus 5 percent. Applicable data shall be recorded at the specified intervals throughout the test. Performance testing shall be conducted with ambient temperature, flow rate, inlet air temperature, and inlet air pressure as specified by the individual tests.

4.7.3.1 Breakthrough and recovery test (type II). The breakthrough test shall be performed at rated flow using saturated inlet air at maximum operating temperature, minimum operating pressure (see 3.7.5) and maximum ambient temperature (see 3.7.1). The unit shall be started in accordance with the applicable technical manual instructions. Dew point readings shall be made using a portable dew point instrument in accordance with MIL-I-24144. By means of manually overriding the cycling mechanism (such as stopping its motor), both towers shall be saturated in sequence in the following manner: with the first tower onstream, override the cycling mechanism so that normal changeover does not occur. Immediately upon breakthrough of the first tower, operator shall switch to the second, having taken precautions against switching to a hot tower. The same shall be done with the second tower and upon its breakthrough the dehydrator shall be returned to its normal cycling sequence. Breakthrough of either tower shall not occur prior to 150 percent of its normal dehydration period. The breakthrough time shall be recorded and the dehydrator shall be operated until the effluent dew point rises to 0°F at a pressure of 80 lb/in^2. Immediately upon breakthrough of the second tower the air outlet valve for the dehydrator shall be closed. The dehydrator shall be operated at this zero flow condition through two complete reactivation periods per tower. Upon completion of the last cooling cycle and subsequent tower switchover, the dehydrator shall then be operated for five cycles at rated flow with air equivalent in quality and

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purity to that of the influent saturated air at maximum temperature and minimum pressure. The dehydrator effluent air shall conform to the requirements of this specification. Any degradation in effluent air quality shall be cause for rejection.

4.7.3.2 Refrigerant leakage test. The type I dehydrator shall be leak tested at a pressure of 140 lb/in^2 on the high pressure side and a pressure of 35 lb/in^2 on the low pressure side, by means of a halide leak detector and shall be free of evidence of leaks with the halide leak detector leak indicator sensitivity of not less than 1/2 ounce per year.

4.7.3.3 Purge flow determination test. Purge flow rate shall be recorded at 80 lb/in^2 and 125 lb/in^2 (see 3.7.4), at zero and rated flow.

4.7.3.4 Electrical functioning test. Switching and cycling actions required for normal dehydrator operation shall be tested at an inlet pressure of 40 lb/in^2 or less.

4.7.3.5 Reliability test. A reliability development growth test (RDGT) shall be conducted for 3000 hours in accordance with MIL-STD-1635 to ensure that the dehydrator shall meet the requirements specified in 3.2. The dehydrator shall be operated at maximum flow with saturated influent air at the maximum operating temperature and minimum operating pressure under maximum ambient conditions. The reliability test shall also demonstrate the dehydrator's ability to meet the requirements of 3.6.8 for loss and restoration of power. The contractor shall determine any immediate and permanent corrective actions for each problem, malfunction and failure. The effectiveness of each corrective action shall then be demonstrated under the same conditions that existed when the original failure occurred.

4.7.3.6 Moisture alarm test (type II). The moisture alarm test shall be performed as follows: Start the unit in accordance with the applicable technical manual instruction. By using the moisture alarm calibration rig and an auxiliary moisture instrument (see 3.6.6.1), the sampling line moisture level shall be raised until it reaches an indicated dew point of 0°F at a pressure of 80 lb/in^2. The dew point and the time difference and the set point at which the alarm is energized and true dew point shall be recorded. Operation shall be returned to normal and it shall be verified that the alarm has de-energized. This alarm test shall be performed five times at different set points equally spaced throughout the dew point range. For test purposes, the first dehydrator of each class shall be calibrated with an ambient temperature such as 120°F to minimize the temperature sensitivity of this type of sensor. Test operations shall not be performed when sensed air temperature exceeds 160°F. Units in production testing shall be calibrated at an ambient temperature compatible with prevailing production test ambient conditions. Equipment shall be marked that the moisture alarm is sensitive to temperature and calibration shall be checked after installation in normal operating ambients.

4.7.3.7 Electromagnetic interference. An electromagnetic interference (emission and susceptibility) test shall be conducted to ensure that the dehydrators conform to MIL-STD-461 (see 3.6.8). Type I and II dehydrators shall meet the test limits specified in MIL-STD-461 for class A4 equipment.

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4.7.4 Maintainability demonstration. A maintainability demonstration shall be conducted in accordance with MIL-STD-471, phase II. The demonstration shall be performed by personnel with the skill level specified in 3.3.

4.7.5 Solid contaminant determination. Independent tests performed on the filter cartridge may be substituted for this test procedure. Should a filter be used which has been successfully tested for this specification in the past, a request for waiver of testing based upon prior acceptance may be submitted. Dehydrator shall meet the requirements specified in 3.6.3.

4.7.6 Inclined operation. The dehydrator shall be operated at rated load for a minimum of one complete cycle per tower for type II dehydrators for each of the four positions and 1 hour for type I dehydrators for each of the four positions. During this period the base shall be tilted 15 degrees in each of the four position conditions: right, left, front and rear. During the test, there shall be no degradation of effluent air quality and all drains shall operate correctly.

4.7.7 Shock and vibration. Upon successful completion of the tests specified in 4.7.1 through 4.7.6, the dehydrator shall be subjected to the type A, high-impact shock tests specified for grade A, class I equipment in MIL-S-901, and for resistance to vibration in accordance with type I in accordance with MIL-STD-167-1. During these tests, the unit shall be energized and the air and water sides pressurized to the nominal operating pressure. When specified in the contract or order, high-impact shock test procedures and a vibration testing report shall be prepared (see 6.2.2).

4.7.8 Operating test. After successful completion of the shock and vibration tests specified in 4.7.7, the dehydrator shall be operated. There shall be no degradation of effluent air quality as specified herein.

4.7.8.1 Type I dehydrator. The type I dehydrator shall be operated at maximum load for 8 hours followed by operation at minimum load for 8 hours. The requirements of this specification shall be met. The transition period shall be not greater than 1 hour. Any operational malfunction, icing of the dehydrator, or malfunction of the automatic drain trap shall be cause for rejection.

4.7.8.2 Type II dehydrator. Type II dehydrator shall be operated with maximum load conditions, maximum temperature and minimum pressure for three complete cycles per tower. Dehydrators shall operate as specified herein. Dew point data shall be taken hourly to meet the effluent air requirements specified in 3.7.5. System pressure shall then be changed to maximum pressure (150 lb/in^2) and all other conditions shall remain unchanged while the unit is operated for two complete cycles per tower. Any operational malfunction of the dehydrator assembly shall be cause for rejection.

4.8 Technical manuals. Technical manual quality assurance management data shall be prepared and delivered as specified in the TMCR and contract or order (see 6.2.2).

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4.3 Inspection of packaging. Sample packages and packs, and the inspection of the preservation, packaging, packing and marking for shipment and storage shall be in accordance with the requirements of section 5 and the documents specified therein.

5. PACKAGING

(The packaging requirements specified herein apply only for direct Government acquisition. For the extent of applicability of the packaging requirements of referenced documents listed in section 2, see 6.7.)

5.1 Packaging. Packaging shall be level A or C as specified (see 6.2.1).

5.1.1 General. Packaging shall be as specified in 5.1.1.1 through 5.1.3.

5.1.1.1 Preservation and packing data. When specified in the contract or order, preservation and packing data shall be prepared (see 6.2.2).

5.1.1.1.1 Packaging, handling, storage, and transportability (PHST) plan. When specified (see 6.2.1), the system or equipment program or acquisitioning manager shall establish the applicable PHST requirements. The plan shall be tailored in accordance with MIL-STD-1367.

5.1.1.2 Lubrication. Rotating joints, bearings, and similar moving parts shall be thoroughly lubricated with the lubricant required for service use. Excess lubricants shall be removed prior to packaging.

5.1.1.3 Painting. Painted surfaces on which the paint is damaged or defective shall be cleaned and repainted with the original specified paint of the same quality and color.

5.1.1.4 Cushioning and wrapping materials. The use of excelsior, newspaper, shredded paper (all types) and similar hydroscopic or nonneutral materials, and all types of loose fill materials for applications such as cushioning, fill, stuffing, and dunnage shall be prohibited. Materials selected for cushioning and wrapping shall be fire-resistant.

5.1.1.5 Asbestos. Asbestos or material and items containing asbestos shall not be used in the packaging of material covered by this specification. When specified in the contract or order, a certificate of compliance shall be prepared (see 6.2.2).

5.1.2 Level A. Each dehydrator unit shall be cleaned, dried, and unit protected in accordance with method I of MIL-P-116 and as follows:

(a) Exposed metal surfaces subject to corrosion shall be preserved with type P-19 preservative, as specified in MIL-P-116.
(b) Internal surfaces or parts normally in contact with water shall be preserved with type P-21 preservative, as specified in MIL-P-116.

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(c) Openings shall be sealed with pipe plugs, plastic caps or plugs, or pressure sensitive, cloth backed, waterproof tape to mil
(d) Throwaway or replaceable filter elements shall be replaced with new filters meeting the requirements of this specification.

5.1.3 Level C. Each dehydrator unit shall be cleaned, dried, and preserved in accordance with ASTM D 3951.

5.2 Packing. Packing shall be level A, B or C, as specified (see 6.2.1).

5.2.1 Level A. Each unit packaged as specified (see 5.1), shall be packed in a container conforming to any one of the following specifications at the option of the contractor:

Specification	Container	Classification
PPP-B-601	Box, wood, cleated-plywood	Overseas
PPP-B-621	Box, wood, nailed and lock-corner	Overseas
MIL-C-52950	Crate, wood, open and covered	Covered, style A or B, type V, demountable
MIL-C-104	Crate, wood, lumber and plywood sheathed, nailed and bolted	Type II, class 2

Container closure and reinforcing shall be in accordance with the applicable container specification or appendix thereto. Anchoring, blocking, bracing, waterproofing and cushioning (see 5.1.1.4) shall be in accordance with MIL-STD-1186. Wood boxes shall not exceed the weight limitations of the applicable box specification. Boxes exceeding 200 pounds gross weight shall be modified by the addition of skids in accordance with the applicable box specification.

5.2.2 Level B. Packing shall be as specified for level A (see 5.2.1), except wood and plywood boxes shall be of the domestic type and waterproofing shall not be required.

5.2.3 Level C. Each dehydrator shall be packed for shipment in accordance with ASTM D 3951.

5.2.3.1 Container modification. Containers exceeding 200 pounds gross weight shall have a minimum of two 3- by 4-inch nominal wood skids, laid flat, which shall support the item and facilitate handling by mechanical handling equipment during transportation, storage and stowage.

5.3 Technical manuals. Two copies of each of the current operation/maintenance technical manuals shall be packaged with each system or equipment. Technical manual packaging shall be as specified (see 6.2.1).

5.4 Marking. In addition to any special marking required (see 6.2.1 or herein), level A, B and C shipment markings, including bar coding, shall be in accordance with MIL-STD-129.

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5.4.1 Special markings. When specified (see 6.2.1), the following markings in accordance with MIL-STD-129 and the applicable container specification or appendix thereto shall be made: structural markings, handling markings such as center of balance, sling and lift points, load bearing and fork lift areas and arrows.

5.5 Unpacking instructions. Unpacking instructions in accordance with MIL-STD-129 shall be placed in a plastic, waterproof bag and sealed. The bag shall be prominently marked "UNPACKING INSTRUCTIONS" and firmly affixed to the outside of the shipping container in a protected location (preferably between the cleats on the container end adjacent to the identification-marking).

6. NOTES

6.1 Intended use. Dehydrators covered by this specification are intended to remove particulate matter, aerosols and water vapor from influent low pressure air and to furnish clean, dry air for use in pressurizing radar equipment, waveguide transmission systems, steam propulsion plant control air systems, operating automatic combustion control and other special low pressure air applications.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type required (see 1.2.1).
(c) Class required (see 1.2.2).
(d) Nominal capacity for class 7 (see 1.2.2).
(e) First article testing, when required (see 3.1).
(f) Mounting, if other than specified (type I) (see 3.4).
(g) If the refrigerant condenser shall be other than sea water cooled (see 3.4.1, 3.6.15 and 3.6.15.1).
(h) Size limitations, if other than specified (see 3.4.4).
(i) Air inlet and outlet design conditions for class 7 dehydrator air capacities other than those specified in classes 3 and 5 (see 1.2.2 and 3.7.5), including:

(1) Inlet supply air pressure.
(2) Inlet supply air temperature.
(3) Outlet air water content (dew point).
(4) Outlet particle size (micrometer rating, see 3.6.3).
(5) Outlet air temperature.
(6) Outlet air flow rate.

(j) Air and water inlet and outlet piping: physical location on dehydrator exterior surface (see 3.8.2).
(k) Whether technical manuals are required (see 3.14 and 5.3).
(l) Inspection conditions, if other than as specified (see 4.3).
(m) Level of preservation and packing required (see 5.1 and 5.2).
(n) PHST plan, if applicable (see 5.1.1.1.1).
(o) Special markings, if required (see 5.4.1).

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6.2.2 Data requirements. When this specification is used in an acquisition and data are required to be delivered, the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (CDRL), incorporated into the contract. When the provisions of DoD FAR Supplement, Part 27, Sub-Part 27.475-1 (DD Form 1423) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Technical manuals (including Technical Repair Standards (TRS) shall be acquired under separate CLINs or subCLINs. Technical manual products (manuscript copy, preliminary copy, reproducible copy, and so forth) shall be developed and delivered as specified in the Technical Manual Contract Requirements (TMCR) exhibits to the contract. CDRLs and DIDs shall not be used for technical manual product acquisitions. Technical manual management data (plans, reports, and so forth) shall be acquired under separate data CLIN, but delivery shall continue to be specified on CDRLs and DIDs. Deliverable data required by this specification are cited in the following paragraphs.

Paragraph no.	Data requirement title	Applicable DID no.	Option
3.5.1 and 5.1.1.5	Certificate of compliance	DI-E-2121	----
3.9	Drawings, engineering and associated lists	DI-E-7031	Level 3
3.10	Imaged aperture/ tabulating cards	DI-E-20477	----
3.11	Failure mode, effects, and criticality analysis report	DI-R-7085	----
3.12	Report, failure/malfunction	UDI-T-23724	----
3.12	Report, contractor engineering and technical services (CETS)	UDI-E-23127	----
3.12	Report, maintainability demonstration	DI-R-7113	----
4.1.2	Inspection system program plan	DI-R-4803	----
4.4.1	Notification of tests	DI-T-23731	----
4.4.1	Report, first article test	UDI-T-23790	----
4.4.1 and 4.5.1	Procedures, test	UDI-T-23732	----
4.7.7	Procedures, high-impact shock test	DI-T-23757	----
4.7.7	Report, vibration testing	UDI-T-23762	----
5.1.1.1	Preservation and packing data	DI-PACK-80120	----

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(Data item descriptions related to this specification, and identified in section 6 will be approved and listed as such in DoD 5010.12-L., AMSDL. Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.2.2.1 The data requirements of 6.2.2 and any task in sections 3, 4, or 5 of this specification required to be performed to meet a data requirement may be waived by the contracting/acquisition activity upon certification by the offeror that identical data were submitted by the offeror and accepted by the Government under a previous contract for identical item acquired to this specification. This does not apply to specific data which may be required for each contract regardless of whether an identical item has been supplied previously (for example, test reports).

6.3 Technical manuals. Technical Manuals Contract Requirements (TMCR) is an exhibit or attachment to a contract specifying technical manual product development and delivery requirements or management data development requirements.

6.4 First article. When a first article inspection is required, the item should be a first article sample. The first article should consist of one unit. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.5 Definitions. The following definitions are included to provide a common basis for dehydrator design and to ensure that the requirements specified herein are interpreted properly:

(a) Dew point or frostpoint. Dew point or frostpoint is the temperature at which a vapor begins to deposit as a liquid or frost.
(b) Cycle. Cycle is the time allowed for a desiccant bed to be in dehydrating service, reactivated, and returned to dehydrating service.
(c) Dehydration period. Dehydration period is the portion of a cycle during which the desiccant bed performs the dehydrating function.
(d) Saturated air. Saturated air is air which, at a given pressure and temperature, contains as much moisture as it can retain in the presence of an excess of moisture.
(e) Breakthrough, type II. Breakthrough is the condition when the effluent air contains more than 0.000015 lbs H_2O/lb dry air, or greater (dew point higher than minus 40°F at a pressure of 80 lb/in^2).
(f) Reactivation period. Reactivation period is the portion of a cycle during which the desiccant bed is heated and then cooled in order to remove the adsorbed moisture.

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(g) Capacity efficiency. Capacity efficiency is the ratio of the effluent air flow to the influent air flow:

$$\frac{\text{ft}^3/\text{min in minus ft}^3/\text{min purge}}{\text{ft}^3/\text{min in}} = \frac{\text{ft}^3/\text{min delivered}}{\text{ft}^3/\text{min in}}$$

(h) Aerosol. Aerosol is a dispersion of solid or liquid particles of microscopic size in air.

(i) Adsorption. Adsorption is the physical process in which water particles adhere to the desiccant (the phenomenon which occurs during the dehydration period).

(j) Maximum load. Maximum load is the set of operational conditions including maximum ambient temperature (120°F), maximum cooling water temperature (95°F), minimum inlet pressure (80 lb/in^2), nominal air flow rate (see 1.2.2), maximum inlet air moisture content for type I dehydrator (saturated at 125°F, that is, 0.015 lb H_2O/lb of dry air), and the maximum inlet moisture content for type II dehydrator (saturated at 105°F, that is, 0.008 lb H_2O/lb of dry air).

(k) Minimum nominal load. Minimum nominal load is the set of operational conditions including minimum environmental temperature (50°F), minimum cooling water temperature (45°F), maximum air inlet pressure (150 lb/in^2), zero air outlet flow, and nominal inlet air moisture content of 0.003 lb H_2O/lb of dry air (saturated at 85°F). For production test purposes, ambient environmental temperature may be used.

(l) Automatic. Automatic implies that no equipment adjustment or manual operations are required for dehydrator startup, operation, or shutdown except the electrical connection and disconnection (by means of an appropriate switch) at startup and shutdown, respectively. Necessary air system piping alignment is not considered as part of the dehydrator operation for the above definition of automatic.

(m) Mean-time-to-repair. Mean-time-to-repair is the sum of corrective maintenance times at any specific level of repair, divided by the total number of failures within an item repaired at each level, during a particular interval under stated conditions.

(n) Preventive maintenance. Preventive maintenance is all actions performed in an attempt to retain an item in specified condition by providing systematic inspection, detection, and prevention of failures.

(o) Corrective maintenance. Corrective maintenance is all actions performed as a result of failure, to restore an item to a specified condition. Corrective maintenance may include any or all of the following steps: localization, isolation, disassembly, interchange, reassembly, alignment and checkout.

(p) Standard cubic feet per minute (ft^3/min). Standard conditions for specified airflows are as follows: 68°F, 36 percent relative humidity (RH), and 14.7 lb/in^2 absolute.

6.6 Provisioning. Provisioning Technical Documentation (PTD), spare parts, and repair parts should be furnished as specified in the contract.

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6.6.1 When ordering spare parts or repair parts for the equipment covered by this specification, the contract should state that such spare parts and repair parts should meet the same requirements and quality assurance provisions as the parts used in the manufacture of the equipment. Packaging for such parts should also be specified.

6.7 Sub-contracted material and parts. The packaging requirements of referenced documents listed in section 2 do not apply when material and parts are acquired by the contractor for incorporation into the equipment and lose their separate identity when the equipment is shipped.

6.8 Subject term (key word) listing.

Desiccant tower
Effluent
Influent
Refrigerated dehydrator
Self-activating

6.9 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Preparing activity:
Navy - SH
(Project 4440-N048)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-D-23523D(SH)	DEHYDRATOR, LOW PRESSURE AIR, FOR SPECIAL APPLICATIONS

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

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